CAUTION

KEEP OUT OF REACH OF CHILDREN
READ SAFETY DIRECTIONS BEFORE OPENING OR USING

Nufarm INTERVENE®

FUNGICIDE

ACTIVE CONSTITUENT:
113 g/kg POLYOXIN-D ZINC SALT

GROUP 19 FUNGICIDE

For the protection of various crops from a range of diseases as per the DIRECTIONS FOR USE.

Certified Allowed Input
ACO CERT. NO. 11540

WG WATER-DISPERSIBLE GRANULE

Nufarm

AUSTRALIAN THROUGH & THROUGH

DIRECTIONS FOR USE

RESTRAINTS

DO NOT apply by aircraft.

DO NOT apply if heavy rains or storms are forecast within 24 hours.

DO NOT irrigate to the point of runoff from the treatment area for at least 24 hours after application.

Spray drift restraints

Specific definitions for terms used in this section of the label can be found at apvma.gov.au/spraydrift

DO NOT allow bystanders to come into contact with the spray cloud.

DO NOT apply in a manner that may cause an unacceptable impact to native vegetation, agricultural crops, landscaped gardens and aquaculture production, or cause contamination of plant or livestock commodities, outside the application site from spray drift. Wherever possible, correctly use application equipment designed to reduce spray drift and apply when the wind direction is away from these sensitive areas.

DO NOT apply unless the wind speed is between three and 20 kilometres per hour at the application site during the time of application.

DO NOT apply if there are hazardous surface temperature inversion conditions present at the application site during the time of application. Surface temperature inversion conditions exist most evenings one to two hours before sunset and persist until one to two hours after sunrise.

DO NOT apply by a boom sprayer unless the following requirements are met:

- spray droplets not smaller than a MEDIUM spray droplet size category

DO NOT apply by a vertical sprayer unless the following requirements are met:

- spray is not directed above the target canopy
- the outside of the sprayer is turned off when turning at the end of rows and when spraying the outer row on each side of the application site

Nufarm INTERVENE® is a contact fungicide with limited mobility. Field performance is reliant on the degree of coverage achieved during application.

CROP	DISEASES	APPLICATION RATE	CRITICAL COMMENTS
Apples	Powdery mildew (*Podosphaera leucotricha*)	22-40 g/100L	Nufarm INTERVENE® should be applied as a protective treatment from flowering through until harvest. **DO NOT** use more than 800 g/ha of Nufarm INTERVENE® per application. Higher use rates are recommended when conditions favour high risk of disease infection. Apply up to 6 applications per season (refer to resistance management guidelines). Reapplication should be conducted on 10 - 14 day intervals. A retreatment interval of 7 days is recommended if rainfall occurs between treatment applications.
	Alternaria leaf and fruit spot (*Alternaria mali*)	40 g/100L	
Almonds	Shot hole (*Wilsonomyces carpophilus*) Suppression only	33-40 g/100L	Nufarm INTERVENE® should be applied as a protective treatment from early flowering (shot hole) OR early hull split (hull rot) through to harvest. **DO NOT** use more than 800 g/ha of Nufarm INTERVENE® per application. Higher use rates are recommended at sites with a history of frequent shot hole or hull rot epidemics, when disease levels were high the previous season or when conditions favour high risk of disease infection. Apply up to 6 applications per season (refer to resistance management guidelines). Reapplication should be conducted on 10 - 14 day intervals. A retreatment interval of 7 days is recommended if rainfall occurs between treatment applications. * or other non-ionic surfactant used in accordance with label instructions for contact fungicides in almonds.
	Hull rot (*Rhizopus* spp., *Botrytis* spp.) Suppression only	33-40 g/100L + 50 mL/100L Nufarm Activator*	
Berries, including: Blackberries Blueberries Raspberries Strawberries (field and protected cropping systems)	Grey mould (*Botrytis cinerea*) Powdery mildew (*Podosphaera aphanis*)	22-40 g /100L OR 220-400 g/ha	Nufarm INTERVENE® should be applied as a protective treatment for fruit, stems, foliage and flowers in line with climatic conditions that favour infection. Higher use rates are recommended when conditions favour high risk of disease infection. Reapplication should be conducted on 7 - 10 day intervals where consecutive treatments are being applied. A retreatment interval of 7 days is recommended if rainfall occurs between treatment applications. Nufarm INTERVENE® can be applied up to 6 times through a growing season (refer to resistance management guidelines). Nufarm INTERVENE® can be applied up to the day of harvest. Pre-harvest applications can assist with reducing disease expression during post-harvest storage.

CROP	DISEASES	APPLICATION RATE	CRITICAL COMMENTS
Cherries (for use in Tasmania and other cool climate regions only)	Grey mould (*Botrytis cinerea*) Suppression only	40 g/100L	Nufarm INTERVENE® should be applied as a protective treatment from early flowering through until harvest. DO NOT use more than 800 g/ha of Nufarm INTERVENE® per application. Apply up to 6 applications per season (refer to resistance management guidelines). Reapplication should be conducted on 10 - 14 day intervals. A retreatment interval of 7 days is recommended if rainfall occurs between treatment applications.
Grapes including: Wine grapes Table grapes Dried fruit	Grey Mould (*Botrytis cinerea*)	40 g/100L	**Production zones frequently experience botrytis expression. Target critical growth stages susceptible to botrytis infection.** Apply Nufarm INTERVENE® as part of a disease management program, prior to the onset of disease. Ensure effective coverage of all surfaces for maximum performance. Apply no more than 4 applications per season (refer to resistance management guidelines). Reapplication should be conducted on 10 - 14 day intervals. A retreatment interval of 7 days is recommended if rainfall occurs between treatment applications.
		22 g/100L	**Production zones with low to moderate frequency of botrytis expression between seasons. Ideally suited to application around sporadic infection events -** Apply no more than 4 applications per season (refer to resistance management guidelines). Reapplication should be conducted on 10 - 14 day intervals. A retreatment interval of 7 days is recommended if rainfall occurs between treatment applications.
	Powdery Mildew (*Erysiphe necator*)	40 g/100L	**Production zones frequently experience powdery mildew expression. Use while severity of infection remains low.** Nufarm INTERVENE® should be applied as a protective treatment. Higher use rates are recommended when conditions favour high risk of disease infection. Apply up to 4 applications per season (refer to resistance management guidelines). Reapplication should be conducted on 10 day intervals. A retreatment interval of 7 days is recommended if rainfall occurs between treatment applications.
		22 g/100L	**Production zones experience low to moderate frequency of powdery mildew expression. Use while severity of infection remains low.** Nufarm INTERVENE® should be applied as a protective treatment. Apply up to 4 applications per season (refer to resistance management guidelines). Reapplication should be conducted on 10 day intervals. A retreatment interval of 7 days is recommended if rainfall occurs between treatment applications.
Basil **Thyme** (protected cropping systems only)	Powdery mildew (*Golovinomyces monardae*)	40 g/100L	Nufarm INTERVENE® should be applied as a protective treatment in line with climatic conditions that favour infection. Apply no more than 4 applications per season (refer to resistance management guidelines). Reapplication should be conducted on 7 - 10 day intervals. A retreatment interval of 7 days is recommended.
	Grey mould (*Botrytis cinerea*) Suppression only	OR 400 g/ha	

NOT TO BE USED FOR ANY PURPOSE, OR IN ANY MANNER, CONTRARY TO THIS LABEL UNLESS AUTHORISED UNDER APPROPRIATE LEGISLATION.

WITHHOLDING PERIODS: NOT REQUIRED WHEN USED AS DIRECTED.

EXPORT OF TREATED PRODUCE

Growers should note that maximum residue limits (MRLs) or import tolerances may not exist in all markets for crops treated with Nufarm INTERVENE®. If you are growing crops for export, please check with Nufarm for the latest information on MRLs and import tolerances before using Nufarm INTERVENE®.

GENERAL INSTRUCTIONS

Nufarm INTERVENE® is a Group 19 fungicide that functions as a chitin synthesis inhibitor, limiting the production of chitin for fungal cell walls. The inhibitory nature of this active constituent creates a need for a variable dose range to match with disease infection load and risk of disease epidemics for different production zones. Select the use rate to match disease risk and infection load based on previous history. Always use as part of protectant program for disease management and select the relevant use rate needed for the growing environment. Nufarm INTERVENE® has limited plant mobility and thus relies on sufficient coverage for maximum disease protection.

Mixing

Nufarm INTERVENE® is a water dispersible granule that mixes readily with water. Three quarter fill the vat with water and initiate the agitation system, then pour in a measured volume of Nufarm INTERVENE®. Ensure that Nufarm INTERVENE® is thoroughly mixed before adding other compatible spray materials. Always consult the correct mixing order for formulations being combined within a single vat. If using in almonds, or if an adjuvant as required by a tank-mix partner, ensure the registered rate of the adjuvant is added to the vat on completion of tank fill. Allow adequate time for the adjuvant to distribute through the vat under agitation prior to initiating field sprays.

Application

Nufarm INTERVENE® is a contact fungicide with limited mobility. Field performance is reliant on the degree of coverage achieved during application. Consideration should be given to the required dilute water volume based on canopy structure and growth stage. Always calculate the dilute water volume for the canopy structure being treated prior to consideration of concentration factors. Calibrate spray units regularly to ensure correct output is being achieved and ensure sprayers are set to effectively target the intended crop zone for maximum coverage. Spray volume requirements, sprayer set up and sprayer operation is likely to change with crop development through the growing season.

Dilute Spraying

Use a sprayer designed to apply high volumes of water up to the point of run-off and matched to the crop being sprayed. Set up and operate the sprayer to achieve even coverage throughout the crop canopy. Apply sufficient water to cover the crop up to the point of run-off. Avoid excessive run-off. The required water volume may be determined by applying different test volumes, using different settings on the sprayer, from industry guidelines or expert advice. Add the amount of product specified in the Directions for Use table. The required dilute spray volume will change, and the sprayer set up and operation may also need to be changed, as the crop grows.

Concentrate Spraying

Use a sprayer designed and set up for concentrate spraying (i.e. a sprayer which applies water volumes less than those required to reach the point of run-off) and matched to the crop being sprayed. Set up and operate the sprayer to achieve even coverage throughout the crop canopy using a minimum water volume of 400 L. **Determine the appropriate dilute spray volume** (See Dilute Spraying above) for the crop canopy. This is needed to calculate the concentrate mixing rate. **DO NOT use at greater than 3X** concentration factors.

The mixing rate for concentrate can then be calculated in the following way:

Example only

1. Dilute spray volume as determined above: for example 1000 L/ha
2. Your chosen concentrate spray volume: for example 500 L/ha
3. The concentration factor in this example is: 2X (i.e. 1000 L ÷ 500 L = 2)
4. Nufarm INTERVENE® use rate of 40 g /100 L is selected. Product is used in 500L/ha water with a concentration factor of 2.
   Vat requirement would be 2 x 40 per 100L or 80 g/100 L of concentrate spray.

Use of concentrate spraying should deliver the same dose rate per ha as dilute applications.

Almonds

Nufarm INTERVENE® provides suppression of hull rot and shot hole in almonds. Nufarm INTERVENE® should be used preventatively from early flowering (shot hole) OR early hull split (hull rot) through to harvest during periods when environmental conditions are conducive to epidemic development. The upper use rates of Nufarm INTERVENE® are recommended for sites that frequently experience shot hole or hull rot epidemics or where disease levels from the previous season were high. Nufarm INTERVENE® is not mobile within the canopy so coverage is critical. Apply sufficient water to wet all surfaces up to the point of run-off.

Apples

Nufarm INTERVENE® provides control of powdery mildew and alternaria fruit/leafspot in apples. Nufarm INTERVENE® should be used preventatively during periods where environmental conditions and canopy growth stages are conducive to epidemic development. Nufarm INTERVENE® is safe for use through flowering and throughout fruit development. The upper use rates of Nufarm INTERVENE® are recommended for sites that frequently experience powdery mildew or alternaria epidemics or where disease levels from the previous season were high.

Berries

Nufarm INTERVENE® provides control of powdery mildew on fruit, stems and foliage, and botrytis on fruit in berry crops. Nufarm INTERVENE® should be used preventatively during periods where environmental conditions and canopy growth are conducive to epidemic development.

Nufarm INTERVENE® is not mobile within the canopy so coverage is critical. Apply sufficient water to wet all surfaces up to the point of run-off.

Cherries

Nufarm INTERVENE® provides suppression of botrytis in cherries. Nufarm INTERVENE® should be used preventatively where environmental conditions and canopy growth stages are conducive to epidemic development. Nufarm INTERVENE® is safe to use from early flowering through to fruit development. Intervene® is not mobile within the canopy so coverage is critical. Apply sufficient water to wet all surfaces up to the point of run-off.

Grapes

Nufarm INTERVENE® provides control of both powdery mildew and botrytis populations in grapevines.

Powdery mildew

Nufarm INTERVENE® should be used preventatively during periods where environmental conditions and canopy growth stages are conducive to epidemic development. Consider historic disease levels when selecting the use rate. The upper use rates of Nufarm INTERVENE® are recommended for sites that frequently experience powdery mildew epidemics or where disease levels from the previous season were high.

Grey mould (Botrytis)
Nufarm INTERVENE® should be used preventatively during periods where environmental conditions are conducive to epidemic development. The upper use rate of Nufarm INTERVENE® is recommended for sites that frequently experience botrytis epidemics or where disease levels from the previous season were high. Nufarm INTERVENE® is suitable for use at many key growth stage timings.
Nufarm INTERVENE® is not mobile within the canopy so coverage is critical. Apply sufficient water to wet all surfaces up to the point of run-off. Air-blast sprayers are recommended for application to vines with very dense foliage.
Export Wine
Consult with Nufarm or the Australian Wine Research Institute regarding acceptable application windows for Nufarm INTERVENE® on fruit intended for export wine production.

Herbs
Nufarm INTERVENE® provides control of powdery mildew and suppression botrytis in basil and thyme. Nufarm INTERVENE® should be used preventatively where environmental conditions are conducive to epidemic development.
Nufarm INTERVENE® is safe to use from early seedling development through to harvest. Intervene® is not mobile within the canopy so coverage is critical. Apply sufficient water to wet all surfaces up to the point of run-off.

Integrated Pest management
Nufarm INTERVENE® is safe for use within integrated pest management programs. Nufarm INTERVENE® shows no impact on yeast species, allowing it to be used safely within integrated disease management strategies with Nufarm Botector® (*Aureobasidium pullulans*).

FUNGICIDE RESISTANCE WARNING

GROUP 19 FUNGICIDE

Nufarm INTERVENE® WG Fungicide contains a Group 19 Fungicide which is considered a moderate risk of resistance development. Avoid applying more than two consecutive applications of Nufarm INTERVENE®, adopting alternation with other functional modes of action where ever possible. Avoid sequential application between seasons in deciduous crop production. Observe suggested restrictions for the number of applications in a specific crop. The number of Nufarm INTERVENE® applications should not exceed 50% of the total fungicide applications within a growing season. Combining Nufarm INTERVENE® with other modes of action that target the same pathogen can assist in minimising selection pressure on the fungicide group. Nufarm INTERVENE® should be used preventatively, prior to onset of disease epidemics. Nufarm INTERVENE® is subject to specific resistance management strategies. For best practise guidelines, refer to resistance management strategies for your relevant crop as published by CropLife Australia.

COMPATIBILITY
Nufarm INTERVENE® is compatible with a variety of commercial fungicides and insecticides. Contact your local Nufarm representative for a detailed account of compatible tank mixture partners.

PRECAUTIONS

Re-entry period
DO NOT allow entry into treated areas until the spray has dried.

PROTECTION OF WILDLIFE, FISH, CRUSTACEANS AND ENVIRONMENT
Very toxic to aquatic life. DO NOT contaminate streams, rivers or waterways with the chemical or used containers.

PROTECTION OF HONEY BEES AND OTHER INSECT POLLINATORS
Moderately toxic to bees. However, the use of this product as directed is not expected to have adverse effects on bees.

STORAGE AND DISPOSAL
Store in the closed, original container in a cool, well-ventilated area. **DO NOT** store for prolonged periods in direct sunlight.
Triple-rinse containers before disposal. Add rinsings to spray tank. **DO NOT** dispose of undiluted chemicals on site. If recycling, replace cap and return clean containers to recycler or designated collection point. If not recycling, break, crush, or puncture and deliver empty packaging to an approved waste management facility. If an approved waste management facility is not available, bury the empty packaging 500 mm below the surface in a disposal pit specifically marked and set up for this purpose, clear of waterways, desirable vegetation and tree roots, in compliance with relevant local, state or territory government regulations. **DO NOT** burn empty containers or product.

SAFETY DIRECTIONS
Will irritate the eyes. Avoid contact with eyes. When opening the container, mixing and loading and preparing spray, wear cotton overalls buttoned to the neck and wrist (or equivalent clothing) and face shield or goggles. If product in eyes, wash it out immediately with water. Wash hands after use. After each day's use, wash face shield or goggles and contaminated clothing.

FIRST AID
If poisoning occurs, contact a doctor or Poisons Information Centre. Phone Australia 13 11 26, New Zealand 0800 764 766.

ADDITIONAL STATEMENTS (WHS REGULATIONS 2011)
Causes eye irritation. Wash hands and exposed skin thoroughly after handling. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present, and easy to do so. Continue rinsing. If eye irritation persists: Get medical advice/attention.

SAFETY DATA SHEET
For further information refer to the Safety Data Sheet (SDS), which can be obtained by scanning the QR code, from your supplier or the Nufarm website.

In case of emergency: Phone 1800 033 498 (24 hrs) and ask for shift supervisor

PRODUCT STEWARDSHIP INFORMATION AND TOOLS
When spraying in or near areas with sensitive crops such as cotton, check online at **satacrop.com.au** for the proximity of sensitive crops mapped by growers.
stewardshipfirst.com.au is a website with CropLife Australia's suite of world-leading product stewardship initiatives, programs and best-practice guides.

CONDITIONS OF SALE
Any provisions or rights under the Competition and Consumer Act 2010 or relevant state legislation which cannot be excluded by those statutes or by law are not intended to be excluded by these conditions of sale. Subject to the foregoing, all warranties, conditions, rights and remedies, expressed or implied under common law, statute or otherwise, in relation to the sale, supply, use or application of this product, are excluded. Nufarm Australia Limited and/or its affiliates ("Nufarm") shall not accept any liability whatsoever (including consequential loss), or howsoever arising (including negligence) for any damage, injury or death connected with the sale, supply, use or application of this product except for liability which cannot be excluded by statute.

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For product advice, or to learn more about complementary Nufarm solutions,
reach out to your local Nufarm specialist or call **1800 NUFARM (1800 683 276)**.

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